Stage 1 & Stage 2 Hard Drill

Chapter 1.4 — Expectation, Variance & Indicator Variables

10 new competition-grade questions, full-chapter coverage. Q1,3,5,7 target Stage 1 speed (under 90s); Q2,4,6,8,9,10 reward Stage 2 depth.

Q1 — Optimal Hedge Ratio

Strategy X has $\mathrm{Var}(X) = 25$; candidate hedge Y has $\mathrm{Var}(Y) = 16$, $\mathrm{Cov}(X, Y) = 10$. You hold $X - hY$ for some $h \in \mathbb{R}$. Find the hedge ratio $h^\star$ that minimizes $\mathrm{Var}(X - hY)$, and the minimized variance itself.
Hint direction: expand $\mathrm{Var}(X - hY)$ as a quadratic in h and complete the square — no calculus required, though either route works.

Q2 — Inverting the Diversification Floor

n equicorrelated assets, each $\mathrm{Var} = \sigma^2 = 0.09$, common pairwise correlation $\rho = 0.25$. Find the smallest n such that $\mathrm{Var}(\text{average})$ is within 10% of its $n \to \infty$ floor $\rho\sigma^2$.
Hint direction: write $\mathrm{Var}(\text{average}) = \frac{\sigma^2[1+(n-1)\rho]}{n}$, express the *excess* over the floor as a single term in n, and solve the resulting inequality.

Q3 — Three-Way Indicator Overlap

Events A, B, C have $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B) = 0.4$, $\mathbb{P}(C) = 0.3$; pairwise intersections $\mathbb{P}(A \cap B) = 0.2$, $\mathbb{P}(A \cap C) = 0.15$, $\mathbb{P}(B \cap C) = 0.1$; triple intersection $\mathbb{P}(A \cap B \cap C) = 0.05$. Let $X = \mathbb{I}_A + \mathbb{I}_B + \mathbb{I}_C$. Find $\mathrm{Var}(X)$.
Hint direction: you need all three pairwise covariances individually (each from its own $\mathbb{P}(\cdot \cap \cdot) - \mathbb{P}(\cdot)\mathbb{P}(\cdot)$) — the triple intersection is a distractor for *this* particular question, not an ingredient.

Q4 — The Second Factorial Moment of Fixed Points

For a uniformly random permutation of $n \geq 2$ elements, let $X =$ # fixed points. Compute $\mathbb{E}[X(X-1)]$ directly from ordered-pair indicators $I_i I_j$ $(i \neq j)$, *without* using the shortcut $\mathrm{Var}(X) = 1$. Then use $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X]$ to re-derive $\mathrm{Var}(X) = 1$ as a check.
Hint direction: $\mathbb{E}[X(X-1)] = \sum_{i \neq j} \mathbb{P}(\pi(i) = i, \pi(j) = j)$ — count the ordered pairs and the per-pair probability separately, the same move as D2's covariance sum but applied to a factorial moment instead.

Q5 — Overlapping Pattern Counts

A fair coin is flipped $n = 15$ times. Find $\mathbb{E}[\#$ occurrences of the pattern HTH$]$, counting overlapping occurrences (e.g. HTHTH contains it twice).
Hint direction: index one indicator per valid starting position (how many are there for a length-3 pattern in n flips?); overlap only affects whether occurrences can coincide, not whether linearity applies.

Q6 — Runs Under Markov Dependence

A coin sequence is *not* i.i.d.: $\mathbb{P}(\mathrm{H}_i \mid \mathrm{H}_{i-1}) = 0.7$, $\mathbb{P}(\mathrm{H}_i \mid \mathrm{T}_{i-1}) = 0.4$, and flip 1 is fair ($\mathbb{P}(\mathrm{H}_1) = 0.5$). Find $\mathbb{E}[\#$ runs$]$ over $n = 20$ flips.
Hint direction: $\mathbb{E}[\#\text{runs}] = 1 + \sum_{i=2}^{n} \mathbb{P}(\text{flip } i \text{ differs from flip } i-1)$ still holds — you need $\mathbb{P}(\text{differ})$ at each step, which requires the *marginal* $\mathbb{P}(\mathrm{H}_{i-1})$ first (does it depend on i, or does it settle to a constant?).

Q7 — Expected Number of Descents

A random permutation of $n = 12$ elements is drawn. A **descent** is an adjacent inversion: a position i with $\pi(i) > \pi(i+1)$. Find $\mathbb{E}[\#$descents$]$.
Hint direction: this is F1's machinery restricted to only the $n-1$ *adjacent* pairs rather than all $\binom{n}{2}$ pairs — the per-pair probability from F1 still applies unchanged.

Q8 — Cross-Inversions Between Two Blocks

A full $n = 52$-card deck is shuffled. Split it into the top $k = 10$ cards and the bottom 42 cards (by final position). Find $\mathbb{E}[\#$cross-inversions$]$: pairs (i, j) with i in the top block, j in the bottom block, $i < j$ (so i comes first), but the card at position i has higher rank than the card at position j.
Hint direction: there are $k(n-k)$ cross pairs, and by the same two-random-values symmetry as F1, each is a disordered pair with probability $\frac{1}{2}$ — the block structure doesn't change the per-pair probability, only which pairs you're summing over.

Q9 — Partial Coupon Collection

A promotion has $n = 40$ distinct coupon types, drawn uniformly with replacement. Find $\mathbb{E}[T_m]$, the expected number of draws to collect *any* $m = 30$ distinct types (stop as soon as 30 distinct types have been seen, not all 40).
Hint direction: G1's independent-geometric-stage decomposition didn't actually use "collect *all* n" anywhere except as the upper limit of the sum — truncate it.

Q10 — Capstone: Non-Uniform Coupon Collector

A market maker interacts with 4 counterparty types per trade, with probabilities $0.4, 0.3, 0.2, 0.1$ (not uniform). Using the inclusion–exclusion identity

$$\mathbb{E}[T] = \sum_{\varnothing \neq S \subseteq \{1,2,3,4\}} (-1)^{|S|+1} \frac{1}{\sum_{i \in S} p_i}$$

find $\mathbb{E}[T]$, the expected number of trades until every counterparty type has appeared at least once. (This is a genuine Stage-2-depth computation, not a 90-second problem — set up all 15 terms carefully before summing.)
Hint direction: group the 15 nonempty subsets by size (4 singletons, 6 pairs, 4 triples, 1 quadruple) to keep the bookkeeping organized, and sanity-check your final number against the uniform case $\mathbb{E}[T] = nH_n$ by asking whether non-uniformity should raise or lower the expected wait versus $4H_4$.

No solutions above by design — Q1,3,5,7 under a 90-second timer (Stage 1 calibration); Q2,4,6,8,9 closed-book with full derivations (Stage 2 calibration); Q10 as a longer sit-down problem. Report which ones took longer than expected — that gap is your next drill target.